Thomas A. Roffe

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Education

Degree

BSc Computer Science @ Uni. of Greenwich
Completed upto 3rd Year

Industry Training

- Red Hat Certified System Administrator 2013→2016
- CCNA Expired
- IRATA Level 1 2014→2016
- Installation & Testing of Class A1 Anchors
- First Aid Course Expired
- Selecting Winners Training
- MBTI Training Myers-Briggs
- HE.NET IPv6 Certified Professional
- Fire Marshall Training Chubb Plc 2012→2014
- QA AWS Solution Architect Training Uncertified

Hoppies & Interests

I enjoy many different aspects of computing, maths, science and technology. I'm currently working on several small personal projects with prototyping and control interface hardware (Arduino and RPis). Additionally, I'm working on creating a Stratum 1 NTP server based on the Raspberry Pi platform using GPS as a time source.

I'm currently a member at a local Weightlifting gym where I regularly go to practise Olympic weightlifting and CrossFit style circuit training and I'm also a keen photographer and garder in my spare time.

Industry Skills & Exposure

Languages & Scripting

Python Go BASH JavaScript HTML CSS Rust

OS & Platforms

Linux Ubuntu Debian Redhat CentOS Gentoo
VMWare ESX ESXi
Windows NT → 11
Cisco IOS CatOS ASA
Nexenta FreeNAS
NetApp OnTap
Apple macOS 10 → 14.4
Fortinet Fortigate Fortimanager
Juniper JunOS
Check Point IPSO
Citrix NetScaler

Software

Services

Apache Nginx MySQL PostgreSQL Exim4 Postfix Courier IMAP OpenLDAP Asterisk iptables Atlassian JIRA Confluence

CICD

Jenkins GitLab Tekton GoCD Concourse CircleCI Github Actions

VPNs

OpenVPN WireGuard OpenSwan Fortinet IPsec Cisco IPsec

Tooling

Terraform Ansible Helm Packer Docker

SCM

Git Perforce Subversion CVS

Cloud Services

AWS

EC2 RDS KMS Lambda AWS SSO SNS SSM Fargate ECS ECR EKS Cloudfront Client VPN Elasticache Cognito GuardDuty SES VPC DirectConnect X-Ray Config Inspector Workspace Certificate Manager Directory Services Directory Services

Google

Bucket Storage IAM Kubernetes Container Registry

Azure

Entra ID Power Automate

Cloud Native Applications

Harbor Hasicorp Vault OAuth2 Proxy Open Policy Agent Keycloak Cilium Multus-CNI Longhorn

Data Science Tooling

Openscoring Hadoop (HDP) HDFS Apache Nifi/Dataflow Jupyter Notebooks

Other

Cat 5 6 6a STP & UTP Cabling
Data Cabinet Cleaning
Refits & Migrations.

Hybirdaccess Technologies Ltd

Present → March 2024

Lead DevOps Engineer

- Architected and deployed a cloud-based backup solution integrated into the application stack for managed customers, leveraging Google Cloud Storage (GCS) and AWS S3. Implemented hourly scheduling for granular data recovery, reducing potential data loss windows by 50%. Automated key and certificate management for encryption at rest and in transit, enhancing security compliance. Developed a containerized Python-based executor, improving scalability and deployment efficiency.
- Collaborated with the Head of DevOps to begin the design and implementation of a Kubernetes Operator, replacing Helm deployments to streamline application management. Leveraged operational expertise to reduce deployment complexity by 30%, improving reliability and maintainability of containerized workloads.
- Initiated discussions on SNAT and CGNAT deployment improvements, driving alignment on compliance and efficiency goals. Developed a Python-based IP SNAT port range mapper to support legal intercept requests and enable deterministic NAT, reducing logging overhead by 40% and laying the groundwork for RFC-4787 implementation.
- Led R&D efforts to implement Cilium and Multus CNIs for multi-NIC containerized traffic server nodes, enabling containerized traffic servers. Reduced latency, improved traffic isolation, enhancing system reliability, security, and management.
- Designed and implemented a multi-tenant Thanos/Prometheus monitoring platform with mTLS-based client authentication, ensuring secure and isolated monitoring for managed customer traffic. Utilized Terraform for infrastructure-as-code and integrated with Helm for seamless client-side product deployment, reducing setup time by 20%.
- Spearheaded CI/CD migration from GoCD to containerized Jenkins, supporting multi-platform (ARM, x86-64) builds for PHP/Laravel, Python, Java, Swift, and C projects. Configured iOS builds using static Mac Mini agents and containerized Android builds, improving build reliability by 35%. Developed a shared Jenkins Groovy library to standardize build patterns, reducing pipeline development time.
- Engineered a Raspberry Pi-based ARM build cluster to support CI/CD pipelines, reducing infrastructure costs.
- Strengthened internal security and supply chain integrity by implementing Git code signing, Gitflow with mandatory pull requests, and application/container signing, reducing unauthorized code risks. Defined initial product specifications for in-toto and Open Policy Agent (OPA) to enhance software supply chain security and policy enforcement. Secured all critical build services with OAuth2 authentication flows, integrated directly or proxied via Azure AD/Entra, improving access control.

Altobyte Ltd

March 2024 → June 2016

BSkyB

May 23 → March 24 (3 months rolling)

Senior DevOps Engineer

- Supported three teams (IOS, Android, Big Screen) BAU operations in all areas of operation for building, testing, signing and release. Manged production crash/debug analytics environments Sentry 9.x, self-hosted GitLab instance, vCenter and associated compute and storage (Sanology). Managed hybrid mix of local and virtual mac and virtual compute (Ask me about it!). LXC containers.
- Migrated on-prem Kubernetes clusters' workloads to Sky's GKE based cloud Kubernetes cluster. On-prem used MetalLB and Ceph. Migrated Helm based applications to Terraform HCL. Including three Jenkins and Sentry.
- Re-write Python 2 application to Python 3 (3.11) and reimplemented application infrastructure storage – PV storage to object-storage based FUSE container driver. Implement authentication and TLS.
- Designed and built a replacement to Jenkins Groovy scripting for release uploads/artifact submission to a custom Sky endpoint. Removing team dependencies on a legacy Jenkins pipeline. Implemented new functionally in teams Gitlab pipeline.
- Sourced, designed and migrated old physical switching infrastructure. Replaced existing 10G switches access switches and refreshed the core/aggregation layer switches. Simplified design due to changing compute requirements.
- Installed new Mac Mini shelfing, hardware (64 M2s). Part refitted and re-cabled data racking around compute needs and lack of HA/Dual homed power/networking on Mac Mini hardware.
- POC evaluation with IOS team evaluating migrating CICD to CircleCI from Self hosted Gitlab. Resulting in the continuation of the GitLab with some
- Interviewed several candidates for DevOps positions within the department. Successfully onboarded two engineers to help support the team ongoing. Worked with Grad team to containerise, secure, and deploy a reporting application. Managed work
- Re-platformed and rearchitected the existing K8 Jenkins CICD platforms for IOS. Migrated Jenkins Master on Virtual Mac OSX Instances to Kubernetes (GKE). Moved local secrets management to Google Secrets Manager. Adapted multibranch and pipeline jobs aligning with Sky's Jenkins Security Policy. Defined, designed and tested job history retention, backup and restore. Improved asset/cache storage. Implemented group-based AD matrix authentication. Automated backup, plugin updates with git versioned Jenkins configuration as code. Security hardening, SSH whitelisting. Moved all agents to web sockets. Prometheus metric exporting. Grafana monitoring dashboard definitions.
- Replaced Jenkins CICD platforms for Android and Big Screen/Desktop Teams using experience gained from IOS team migration. Additionally worked with the teams to adapt to the new environments, including adapting pipelines, creating custom build containers, adapted signing operations.

HMRC / Equal Experts

August 22 → February 23 (12 months)

Platform Security Engineer

- Performing various platform risk audits and security assessments. Enhance and improve existing platform tooling for alerting and compliance. for example HMRCs hmrc/platsec-compliance-alerting and hmrc/platsec-aws-scanner
- Update and improve existing CI/CD pipeline to include better pre-release testing coverage and security scanning.
- On-call and platform security support duties on a rotating basis.

HMRC / Equal Experts

Jan 22 → August 22 (12 months)

DevOps Engineer

- DevOps engineering for 5 internal HMRC products.
- Built CI/CD pipelines for a number of mature projects that lacked release pipelines.
- Reengineered a number of internal HMRC terraform modules to better align infrastructure with HMRCs security standards. Introduced policy-as-code testing framework with custom rules via Checkov.
- Improved secrets handling via Hashicorp Vault

BP / CloudIgnite

April 21 → Oct 21 (3 months rolling)

DevSecOps Engineer

- Assist, Administer, Operate and Develop BP's AWS central logging services. Aggregating service logs and instance logs from ~900 accounts into various object stores by product offering, inline with CloudWatch, VPCFlow Logs, Inventory, Config and CloudTrail services.
- Re-designed and developed "account provisioning" service, which was responsible for provisioning, deploying and configuring new account services and IAM least privilege access rights to new or existing accounts to enable collection of logs cross-account and cross-region into centralised buckets. Ground up rewrite in OOB Python. Provisioned new account bucket permissions and dealt with oversize IAM policy behaviour, provisioning new CloudWatch log destinations and regional kinesis firehose (CloudWatch log) streams.

Hitachi EU

Dec 20 → April 21 (4 months + 1 month extension)

Senior DevOps Engineer

- Productionise a POC for EV (Electric Vehicle) Fleet Management and Decarbonation Platform. Transforming a small development environment on a single EC2/Docker host to a full infrastructure design and implementation with build, test and release pipelines and dependent services for over 19 microservices running on a diverse and fault tolerant ECS environment. Re-platforming users and services to new development, UAT and production environments while progressing system development in parallel.
- Further developed Terraform modules and HCL definitions from initial project state (POC) to a multi-layered, multi-tenant application stack in multi-environment IaC deployments.
- refactored
- Help develop application authentication flow using AWS Cognito User Pools. Extensive use of Terraform for provisioning tenant/client user pools called programmatically as a part of new customer signing logic.
- Developed AWS Pipeline MS teams alerting application. Enabling message filtering and dedupe of AWS Pipeline state alert messages. Developed in Python 3.6 and packaged for AWS Lambda functions with corresponding build, package and development pipelines. Utilising AWS CodeDeploy blue/green deployments.
- Deployed AWS SSO (Single Sign-on) into a single org. account and Integrated SSO into each subordinate org accounts for the business group.
- Designed and deployed AWS Client VPN with AWS SSO (OAuth2) backing. Implemented robust security and access controls with group based access controls. Implermemented Cross Account VPC peering and associated security controls.
- Delivered e2e test suite/harness (Karate). Worked closely with test engineers to implement a schedule based test automation platform. Implemented test automation reporting and alerting; delivering failed tests reports to designated developers and test engineers.
- Designed and partially implemented initial monitoring and alerting services using Prometheus, Alert-Manager and Push Gateway. Metrics and graph visualisations via Grafana.

Ti-Media Ltd / Future Plc

Nov 19 → August 20 (3 months rolling)

Senior DevOps Engineer

- Lead architectural and design steering group for infrastructure team together with the development team for a number of internal system projects to upgrade and update the underlying infrastructure. including some of the development tooling with regards to build, test and release strategies on a number of publications.
- Deliver POC for Build, Test, Release and Deploy of TI's subscription ecommerce platform. Platform built with AWS codepipeline, terraform, ECS with EC2 (Originally Fargate) with associated GitOps tooling for build, test and release pipelines to application containers repos.
- Designed, built and delivered site/content image optimization service which reduced our CDN and traffic transfer cost by approx 30% for non-apple based clients. Image Converters Service allowed for delivery of images encoded in webp for supported browser types, with cdn/cloudfront caching and fallback handling to content origin. Originally designed and written as a Python/Lambda but reimplemented a Flask API due a number of technical limitations integrating webp libraries and size limitation of returned objects.
- Containerization of new Decanter Magazine Wine Review Node application. Assisting development team integration with new containers into the existing application stack and existing Jenkins pipelines. Provisioned DynamoDB with associated terraform.
- Assisted Marie Claire development team with initial Lambda deployment and subsequent performance troubleshooting and tuning. This project enabled parallel consumption of supplier feed from approx. 30 large fashion brands, ultimately significantly reducing time to market from initial announcement of product on supplier feed to
- Integrated X-ray log and event trace handling into a number of hard to debug application to better assist Image Converters Service - webp for supported browser types, with cdn/cloudfront caching and failback handling.
- IAM policy and account audit for all TI's AWS accounts, consolidated into AWS IAM Organization for improved policy reporting and enforcement. Integrated Cisco CloudWatch into all org. accounts.
- Assist team with production incidents and troubleshooting as BAU operations as and when required.Additionally started project planning and initiated terraform layout and repos for POC using EKS for greater service improvement among other things associated with better container management and resource utilisation with Kerbernates.
- Assisting the legal department with content removal and media from TI's media library which lead to developing a Python API for removing multiple CDNs cache content and origin source media from TI publication media silos for when legally required/requested.
- Various administrational Cloudfront and Fastly CDN management tasks and new deployments. Automated repetitive tasks some with Jenkins and Terraform pipelines for fun and profit.

British Medical Journal

June 19 → August 19 (3 month + 1 month extension)

Senior DevOps Engineer

- Lead design and implementation of rollout of Remove Access / OpenVPN solution based on AWS Client VPN Services. Designed and configured VPC networking, routing, security into existing deployment. Utilised AWS ACM certificate & Managed Active Directory for user authentication and group based authorisation controls. Managed initial rollout to key stakeholders in initial testing and troubleshooting. Worked with inhouse IT support services to roll out OpenVPN application on managed desktops. Integrated VPN networking into existing inhouse AWS VPC Terraform codebase.
- Extensive work with BMJ Lead developer to develop the design for BMj's async streaming application in Java spring boot using AWS Kinesis, Lambda and RDS services. Encouraged use of AWS secrets manager for application secrets. Infrastructure implementation in Terraform. Automated application build and deployment pipeline. Performance and Acceptance testing.
- Jenkins Build Pipeline Improvements, Operations infrastructure focused. Implement Terraform in Docker. Refactor some existing inhouse terraform modules. Move to IAM role based Terraform. Extensive security harding of Jenkins hosts.
- AMI image creation with packer, bootstrapping base images based on role for faster deployment with view to start using ASGs. implemented prerequisite AWS infrastructure defined in Terraform with fully automated build, bootstrap, deploy and share, use mechanism.
- Lead group training for terraform and platform support. Training team on use of newly implemented terraform in docker jenkins pipeline pattern. Support developers and operations team members with Terraform tooling and AWS usage in general.
- Extensive solution, process and tooling documentation for the various new platforms. for example user management process for client VPN.
- General Application and Operation support of various inhouse applications.

OpenBanking

November 18 → March 19 (3 months rolling)

DevOps Engineer

- Terraform IaC built on AWS EC2, RDS, Certificate Manager, KMS, Directory Services, Lambda, SNS, Config, Inspector, SSM, ECS & ECR.
- Produce and develop docker images within Jenkins pipeline for building various internal and externally hosted applications. Predominantly Python and Node Applications.
- Rearchitected Terraform codebase to allow more consistent use of shared infrastructure variables, be more DRY and promoted and migrated the use of terraform modules.
- Central Resource for development team, ensuring effective productionisation of newly developed applications. Ensuring initial capacity, fail-over/fault tolerance method and metrics are captured and recorded, monitored and actioned, monitoring methods, support and deployment playbooks. etc
- Enabled more effective methods for developers to test and build development docker images on AWS infrastructure.
- Packer AMI build & Ansible refactoring, Performance improvements. Build was taking over an hour, reduced with role specific AMI builds and pre-installed system pre-requisites.
- Cost Control: AWS Lambda Function to powered down development infrastructure overnight and weekends.

Camelot Group

June 2017 → October 2018 (3 Months Rolling)

Senior DevOps Engineer

- Terraform IaC built on AWS EC2, RDS, Certificate Manager, KMS, Directory Services, VPC Direct Connect, Lambda, SNS, Config, Inspector, SSM, AWS Workspaces.
- Part designed and implemented Terraform and Ansible code for the deployment of Infor technologies OCCM and IA products (Java Application with MS SQL backend and HTTP and SMTP public facing services) Single region, multi-AZ deployment. Implemented HLD to POC and Production. Designed and implemented basic build, test, deploy pipeline for future expansion and product upgrades utilising three separate AWS accounts per environment (dev, test, prod).
- Championed data siloing and least privilege separation for AAA services. covering Windows AD and interdomain trust relationships for both data and application teams.
- Engineered TrendMicro Deep Security Manager 10 Terraform module for deployment and upgrade of DSM products/virtual devices. Design, tested and deployed DSMs to AWS and enrolled approx. 250 mixed server environments. Agent deployment for Linux hosts was performed with a custom Ansible playbook and windows hosts done via AWS SSM and Windows Powershell. Future versions were baked into both Windows and Linux AMIs via custom packer builds.
- Built Terraform and CloudFormation integration for part automating the deployment AWS Direct Connect peering into new accounts.
- Built Terraform modules for CIS 2 compliance checking via AWS Config and filtering and alerting via AWS SNS and Lambda/python. alerts pushed via ChatOps (Google Chat/Hangouts)
- Git Repo management; code convergence strategies testing plans, pre-commit checking, repo branch locking and security and contribution notifications via ChatOps/webhooks. Common system and application roles had Ansible playbooks rolling into AMI Packer builds.
- Instrumental in adding Ansible playbook InSpec testing via Vagrant along with code linting to catch the basic errors when writing ansible playbooks.
- Other tasks, duties and points of note; Operations and Production support duties. Helped select and grow the CloudOps teams from 2 people (self included) to 5 engineers (2 contract, 3 permanent). Attained multiple contract renewals over multiple projects

ASOS

June 2017 → July 2017 (1 Month Rolling)

Linux System Administrator

- Part-automated installation of or Oracle Java and Oracle Retail Stack using Ansible. OS to APP
- Implementation plan for FreeIPA configuration for DNS, Kerberos and User Management role out.
- Supported Oracle Retail Developers on JVM tuning and batch process scheduling and monitoring and alerting.
- OS User and resource management

Bytesumo Ltd

May 2017 (Direct Consulting)

DevOps Engineer

- Onsite Hortonworks HDP Cluster buildout - OS Install and HDP Configuration on 3 x Mac G5s
- OpenVPN Configuration for remote access
- FreeIPA configuration for DNS, Kerberos and User Management
- Python, Anaconda Scale, PySpark and Jupyter Setup and Configuration
- Installation of Spark, TensorFlow, Torch7, Lua and Associated ML toolsets
- Developed Ansible Playbooks for to bootstrap instances for HDP and Ambari Install, Ensuring all hosts meet the HDP OS requirements,
- Migrated Large Dataset to new HDFS Cluster
- Configure new Cisco Switching Infrastructure

M4 Data

May 2017 (Direct Consulting)

DevOps Engineer

- AWS Hortonworks HDP cluster buildout on AWS EC2 Instances and Associated VPC, S3 Bucket Configuration.
- AWS IAM User creation and management
- S3 Bucket Configuration for Data Operations
- Load RAW datasets into S3 for Data Science Team to consume.
- Sharing Results Data from the Data Science Team with End Customers in a secure and encrypted manner.
- Security Review of Existing AWs infrastructure. implemented standard recommendations from amazon as a base policy.

HMRC

July 2016 → April 2017 (3 months rolling)

DevOps Engineer

- Managed Github Enterprise Services, Atlassian Jira and Confluence Service for 5k users and developers of HMRC Digital teams nationally.
- Managed backend support system Deskpro, supporting internal HMRC Phone agents and 2nd and 3rd line operation for HMRC Help and Support services for the rate paying public thought tax.service.gov.uk services.
- Experience of managing public facing systems in constant use with peak load use cases. Successful implemented performance changes prior to self assessment peak. Including web server php-fpm performance tuning, mysql database turning. Performance validation with various load and user journey tests both pre and post changes.
- Support various database replication and backup scenarios in use throughout the tooling department.
- End of Life atlassian confluence upgrades and jira upgrade for 6.4 to 7.2
- Continuous Upgrade of Deskpro Support system, enabled though better automation. Development of Ansible playbooks to enable more predictable upgrades.
- Migration Project for moving current Puppet CM to Ansible and assisted roles and plays.
- Development of tooling estate dashboard (aka Doomsday Clock) for current licensing levels for google apps, atlassian products and plugins and deskpro written in python using falcon framework on the backend and static/javascript frontend. This allowed for data to be pushed into Grafana to enable management to better understand the trends in user licensing throughout the year. (manly due to high turnover of staff)
- Personally responsible for £120K licensing saving for Deskpro. Using a mix of python and sql queries I successfully removed 1200 unused licences that were to be erroneously renewed.

KPMG UK LLP Ltd

June 2016 → June 2015

DevOps Engineer

- Managed our on-prem Hadoop cluster and oversaw the addition of a new host to the cluster. General user management, YARN/SPARK Job management, HDFS tuning and data ingest tasks.
- AWS VPC / IPA OpenVPN IPTables integration work. Enabling group based network access controls using the
- OpenVPN/IPA/ AWS VPC APIs using python via the scripting functions in OpenVPN
- A large proportion of my duties involved migrating away from puppet and bash scripts to Ansible for automating the bootstrapping and development of applications and services.
- Work effectively with team members to develop Ora (an internal inventory, compliance and orchestration tool) this enabled other teams within the firm to spin up hadoop cluster(s) (with additional data science tooling) on AWS on a per-project basis via a web interface.
- AWS training; Solution Architect course, Ansiblefest
- Jenkins/Packer Builds and Deployment for AWS images
- Instrumental in packaging, designing and deploying Apache Nifi installation to enable multi stream, massively parallel data collection in near real-time.
- Implemented Elasticsearch stack with Logstash and Kibana for log visualisations.
- Leading resource on bringing Openscoring into Production within KPMG. Built RPM packages, Designed the workflow and tooling for effective model deployment. Worked on log monitoring and reporting for billing purposes. Contributed to the Openscoring project and inspired the design of the python Openscoring module.
- Built the Data Science and Engineering Toolbox images based on CentOS7 for a number of visualisation (VMWare, Xen, Virtualbox, AWS) platforms to facilitate the Data Science team to work remotely on client sites and have a full complement of Data Science tooling at hand in the situations that we was prevented from using our cloud or on premises cluster for data security and data governance reasons.
- Worked with compliance offers to ensure best practice in data and system security ensuring data safeguards, audit logging and accounting was robust and accurate in accordance with best practice data security methodologies and KPMG policies. data removal was done in an audited and reproducible in an automated manner.
- Implemented Full Disk Encryption on our existing cluster and automated future deployment of secure hosts utilising LUKS and Ansible.

Qinectic Ltd

June 2015 → October 2014

DevOps Engineer

- Build and design of HA Monitoring, Logging and Analysis platform build on Sensu, logstash, Kibana, elasticsearch, RabbitMQ
- Extensive orchestration scripting for the deployment, installation and management of various servers hosted on AWS, Rackspace and on dedicated hardware at Colo DC
- Knowledge of working with N3 (NHS National Network). Change control and data and security policies.
- Designed, built and implemented RabbitMQ Cluster for application message queues and logging for Sensu monitoring clients.
- Build, Deploy and Manage Custom applications packages for Debian (deb's) and Redhat (rpm's) distributions and repository management with PKI for package signing.
- Deployment a number of virtual and physical machines with a number of hosting providers. Initial configuration and bootstrap using Ansible.
- Built custom AWS AMIs for faster deployment of new internal infrastructure and customer solutions
- Escalation point for customer application support. Including supporting third party customer integrations though the application API and related tools.
- Upgrades of hosted PHP application. Including migrations from older versions of the codebase to newer packaged versions. Upgrade of customer MySQL database schemas ensuring backup and roll-back procedures are still valid using a manual process or Doctrine
- Internal Desktop and Network Support including the redeployment of MS Windows AD server (2008R2) from Windows SBS
- Designed and Built a developer self-service platform for live database export and obfuscation for using medical records in test or development environments using Ansible and Ruby Strike and a PHP front end developed on Symphony

Claranet

October 2014 → August 2014

Linux Hosting Implementation Engineer

- Build out of customer solutions from predefined service offerings designed by solution architects.
- Operated across a wide and diverse set of data centres in various countries and to different designs and standards.
- 3rd Line escalation point for hosted customers and a part of the On Call team.
- Deployed a number of virtual hosting platforms with a mix of Operating systems.
- Deployment of new customers to F5 and HAProxy load balancers.
- Customer first training (third party customer focused training)
- Designed and Built PXE boot environment for the automatic installation of Linux Operating Systems utilising standard UNIX tool chain. iPXE, TFTP, DHCP and a lightweight PHP front end.
- Scripting; automation and orchestration using Ansible for the deployment of various technologies. Nginx, HAProxy, Apache, MySQL, Redis and others.

NTT Europe

June 2014 → December 2011

Deployment Engineer

- This role was wide and varied in terms of responsibilities which included BAU tasks listed below, assisting in the sales & pre-sales process to ensure designs fulfil both customer requirements and operational efficiency goals, technical customer consultations (on and off-site), customer fault analysis and reporting.
- Managed and implemented new customer's installations and migrations if required for customers with or without project management or technical architectural (TA) resources. Being a technical authority for small/medium customers after going live or as a COE being led by a technical authority for large enterprise customers.
- Stand up, configuration and migration of customers on shared and/or dedicated load balancing services from Alteon or Netscalers devices to newer Netscalers (EOL-Migrations)
- Myself and the Deployment team were pivotal in migrating over 100+ customers from a central London data centre to a new facility in Hemel Hempstead.
- Stand up, configuration and migration of customers on shared and/or dedicated load balancing services from Alteon or Netscaler devices to newer Netscalers (EOL-Migrations)
- Installation of various firewall hardware devices from a number of vendors Fortinet, Cisco, Juniper, Checkpoint and software firewalls like IPTables and Windows FW. Undertaking everything from basic configuration, to policy addition and creation, VPN creation and management, AV + IDS/IPS configuration & migrations. Configuration and management of routing tables and protocols (RIP, OSPF) in build and production.
- Ensure quality assurance tasks are completed on all installed systems ensuring the audit trail is kept up to date and accurate throughout the project life-cycle. Ensuring best practice is maintained in terms of high availability and security.
- Contribute towards the improvement of deployment processes. Feeding back feature requests and bug reports to the OSS development teams and the Systems Engineering teams.
- Coordinate and oversee tasks with other NTTE personnel worldwide for customer migrations at remote DCs and out-of hours work and maintenance. Additionally occasional travel was required to support project implementation or migrations.
- Delegate tasks to outsourced teams in the various EU Data Centres. For Example Cabling Backup Tape Rotation.
- Maintenance of technical competencies across a wide ranging set of Internet technologies with a very keen and eager attitude to learn new technical subjects within computing or not.
- Employ soft skills to assist in creating a positive experience for the customer.

Virtual Internet Ltd

November 2011 → February 2011

Systems Engineer

- Supported customers of various sizes from Government Departments, Large Enterprises, SME's to Individuals under different SLA levels for managed and unmanaged services.
- Typical Working Week with On-Call & Out of hours working patterns. Variety of work from simple password word resets and domain renewals to complex project work with multiple servers, firewalls switches, sans etc.
- Provisioning VMware Hosts – ESX1 4.1 vCloud Director, vSphere – Managing, Installing, Troubleshooting ESXi installations
- Physical Server Provisioning with OS and Software stack configurations on various Windows and Linux platforms.
- Provision of VMware Cloud Servers and OnApp (Zen) Cloud Servers including custom builds and from templates. Specialized in Hybrid Cloud Implementations.
- Support customers of remote data centres (Manchester). Liaising with colleges to affect changes such as physical adds, moves and installs etc. Organization of shipping and handling of servers in and out of DC.
- Planning of initial phase of a server migration project for ~200 Servers from Manchester to new London data centre.
- In-house & Customer SAN installation and support duties using Nexenta. Including snapshot management, licensing reporting, disk expansion, status & health monitoring.
- Backups system administration using NetVault including patching, library size expansion, SAN expansion, client support and installation and basic housekeeping duties
- Sourced, built and configured and administered IP PBX (VoIP) server with Cisco handsets. CLID Lookup with Ticketing System Integration (OTRS) built on Asterisk Phone System/FreePBX
- In-house and customer system and network monitoring using Zabbix and Cacti
- Administered and supported a HA Apache platform for High Availability customers with multiple MySQL replicated instances.
- Administered and supported multiple POP and SMTP servers for a large number of POP Mailbox customers
- Administered and supported Hosted Exchange Platform
- Documentation writing – For customer projects and internal infrastructure and systems
- Regular physical DC security/stock audits, power usage reporting duties
- Installation, Configuration Management and support of Cisco Switches, Routers and Firewalls (ASA & Pix) with IPS Security modules. Managed multiple instances of IPTables and pfSense for customer solutions with varying needs, size and traffic requirements.

VPS.NET

December 2010

Contract Systems Engineer

- 1Day Installation of 2 Supermicro based SAN's running StarWind on Windows 2K8 in Amsterdam, Netherlands at the EvoSwitch Data Center.

Central Height Safety Ltd

February 2011 → July 2009

Systems Engineer / Operations Manager

- Designed, Built and hosted company website http://www.centralheightsafety.com/. Built on Linux, PHP Zend Framework, JQuery and MySQL. Designed XHTML/CSS Styling to comply with company brand.
- Designed and Built and sourced network infrastructure from scratch built on Windows 2003 Server and Debian Linux with Windows XP, Mac OSX clients (some Linux desktops). Installed and configured Switches, Routers and VoIP PBX (Asterisk), and an OpenVPN implementation. Backups and MS Exchange.
- Analysed, designed and implemented custom CRM, Work bill, job scheduling and asset tracking platform built on Zend Framework (PHP), JQuery and MySQL (with replication) running on Linux Debian 5 (Lenny). With multiple front ends for field service staff via their IPhone and administration staff via a web browser.
- Single port of call for IT support issues for end user support and server issues. Data Protection and Security duties. Supported a total of 5 Servers (2 Windows 2K3, 3 Linux Debian Systems)
- Analysed, Sourced & Implemented Time and Attendance System
- Started implementing an e-commerce site for equipment sales. (Shelved due to budgetary & time constraints).
- Managed and Implemented successful migration from Microsoft Exchange 2003 to Google Apps Service with an 85% reduction in operating and maintenance cost.
- Designed and built internal web database application for tracking of staff and training equipment and retail/sold equipment to comply with Health and Safety at Height Laws (HSE) and IRATA.
- Managed marketing campaigns, budgets and production of digital and printed media.
- * This role was very wide and varied; I've omitted certain roles and responsibilities for sake of keeping my CV relevant to the industry. But I'm more willing to provide a full description of my duties on request.

Travelling

July 2009 → April 2009

- Travelled around various countries in Europe for four months.

Perforce Software Ltd

April 2009 → March 2008

Support Engineer

- Operated within a global team that supported the client/server application to development teams of all sizes. From 1 to 10,000 users from national to international companies and government agencies, often working under strict NDA agreements.
- Specialized in Operating System integration (mainly Linux) though external tools such as log rotate, backup manager, p4ctl and monitoring through custom Perl scripts and RRDtool.
- Specialized in remote access such as VPN, SSH set-ups and best practices for customers need to comply with specific regulations and accounting I.e. Sarbanes-Oxley, military export laws
- Advised customers on performance tuning and best practice set-ups and configuration. Supported integrations, branch, merge and release cycles for small and simple developments to the large and complex

City Central Ltd

February 2008 → December 2007

Contracted IT Consultant

- Upgraded server hardware to single Quad Processor HP ML350 with a RAID5 configuration and LTO3 backup tape drive.
- Migrated Windows 2000 to Windows 2k3, Active Directory, Users & Data.
- Upgraded all software - Office XP to 2k3, Sage 50 v10 to v12, Industry package CleanLink, setup and integrated with Windows CE PDAs and synced with outsourced MS Exchange.
- Implemented a Backup and DR strategy + server monitoring and reporting, centralized AV - Upgraded all workstations to Windows XP from Windows 2000.
- Upgraded Switching and Routing fabric + Implemented VPN
- Equipment and cabinet clean, Wire tidy/colour coding, UPS testing + set-up. - Migrated Email from MDaemon mail server to MS Exchange.

Reporo Ltd

December 2007 → February 2007

Infrastructure, Support & Test Engineer

- Reporo provides off portal mobile social networking services to in house brands (Reporo.com & Mwah.com) and third parties like MyMobileLife (Carphone Warehouse)
- QA / Testing on Java enabled handsets and emulators
- Bug Tracking, Reporting & Management via Mantis Bug Tracker
- Installation, Maintenance and Monitoring of production and development environments ensuring PCI (Payment Card Industry) compliance.
- Fully supported application stack built on Apache 1.3, Java, Perl, PHP and MySQL and Linux (Debian Etch)
- Fully administered Linux iptables firewalls and Checkpoint firewall. Managing and implementing NAT, Policy Rules, routing tables.
- Implemented Performance, configuration and security monitoring through the open source HyperIC Monitoring platform and an Office VPN solution (OpenVPN)
- Designed and Implemented PHP based CMS for internal use to allow publishing of copy and images, which used patterns like MVC and some AJAX
- Managed Office Move and Interior refit with builders and installed new CAT5 cabling and Data Cabinet Refits

Dow Jones Newswires

November 2006 → August 2006

Systems Support Analyst

- Experience in corporate methods and formal processes.
- Provided 1st Line Telephone support for EMEA, Asia and Africa
- All tasks and responsibilities undertaken followed highly detailed protocol for logging and reporting all projects and faults
- Precision time management with regular reports to management - Use of Remedy Fault Ticketing System.
- Support of Internal Bespoke Application as well as standard MS Office Packages
- Supported all workstation hardware and software – Apple, Dell Hardware – OSX and Windows XP Operating Systems (Some Solaris on Sun Hardware)
- Supported All Mac OSX users in the UK, with 3rd line support provided from the US

PaceFix Ltd*

August 2006 → August 2005

Director

- More information available on request.

Institute of Education, UCL, EPPE Project*

August 2005 → June 2005

Temporary Data Entry Role

- More information available on request.

Think Computers Ltd*

August 2005 → February 2004

Contracted IT Support Engineer

- More information available on request.

London Borough of Southwark

September 2003 → July 2002

Field IT Support Engineer

- More information available on request.

* Positions held during university studies